matrixH.C Options Summary

Signature:
```
matrixH(TString trainAO2D = "AO2D.root", bool useAO2DTraining = false, TString dataAO2D =
"AO2Ddata.root", TString truthAO2D = "AO2Dtruth.root", int useAO2Data = 0, int priorMode =
0, bool subtractFakeFeed = true, double centMin = 0.0, double centMax = 100.0)
```

Main Switches

trainAO2D	MC training file. Used for the response matrix when useAO2DTraining = true.
useAO2DTraining	false: generate toy response. true: read response/training from the MC AO2D table O2npmcchargedtabl.
dataAO2D	Reco-data-like input. For real data this is read from O2npcollisiontabl plus O2nprecochargedca.
truthAO2D	Truth input for independent-MC closure when useAO2Data = 3.
useAO2Data	0: generated pseudo-data. 1: real reco data. 2: RecoTrain/TruthTrain same-sample closure. 3: independent MC2 reco-table plus MC2 truth closure. 4: MC2 reco and truth both read from O2npmcchargedtabl.
priorMode	0: flat positive prior. 1: TruthTrain fixed-point/debug prior. 2: RecoTrain-shaped prior. 3: Poisson-randomized TruthTrain prior. 4: Poisson-randomized flat positive prior.
subtractFakeFeed	true: subtract fake/feed-in estimate from reco input before unfolding. false: bypass fake/feed subtraction.
centMin/centMax	FT0M centrality percentile interval applied only when reading data-like reco collision tables O2npcollisiontabl. Default 0-100 keeps all collisions.

Useful Output Histograms

hUnfold	Final unfolded 2D result in pt and multiplicity.
hUnfoldPtProj	pt projection of hUnfold.
hUnfoldMultProj	track-weighted multiplicity projection of hUnfold.
hUnfoldMultEventProj	hUnfoldMultProj divided bin-by-bin by multiplicity M; approximate event-style multiplicity view.
hTruthDataMultEventProj	hTruthDataMultProj divided bin-by-bin by multiplicity M; closure reference for hUnfoldMultEventProj.
hTruthTrainMultEventProj	hTruthTrainMultProj divided bin-by-bin by multiplicity M; same-sample/debug reference.
hRatioPt	hUnfoldPtProj / hTruthDataPtProj.
hRatioMult	hUnfoldMultProj / hTruthDataMultProj.
hRatioMultEvent	hUnfoldMultEventProj / hTruthDataMultEventProj.

Closure / Data Modes

1. Toy Response + Toy Data

Switches	useAO2DTraining = false, useAO2Data = 0
Meaning	Training response is generated by the toy model. Pseudo-data is generated independently by the same toy model.
Compare	hUnfold to hTruthData. Projections: hUnfoldPtProj to hTruthDataPtProj, hUnfoldMultProj to hTruthDataMultProj, hUnfoldMultEventProj to hTruthDataMultEventProj. Ratios: hRatioPt, hRatioMult, hRatioMultEvent.
Example	`matrixH("AO2D.root", false, "AO2Ddata.root", "AO2Dtruth.root", 0, 0, true)`

Closure / Data Modes

2. Same-Sample Closure / Debug

Switches	useAO2Data = 2. useAO2DTraining can be false for toy response or true for AO2D MC response.
Meaning	RecoData is replaced with RecoTrain and TruthData with TruthTrain.
Compare	hUnfold to hTruthTrain. Projections: hUnfoldPtProj to hTruthTrainPtProj, hUnfoldMultProj to hTruthTrainMultProj, hUnfoldMultEventProj to hTruthTrainMultEventProj. With priorMode = 1 this is the fixed-point/debug check.
Example	`matrixH("AO2D.root", true, "AO2Ddata.root", "AO2Dtruth.root", 2, 1, false)`

Closure / Data Modes

3. MC Response + Real Data

Switches	useAO2DTraining = true, useAO2Data = 1
Meaning	trainAO2D provides the MC response/training. dataAO2D provides real reco data. truthAO2D is not used. centMin/centMax select the FT0M centrality percentile range in O2npcollisiontabl.
Compare	For real data there is no truth closure target. Inspect hUnfold, hUnfoldPtProj, hUnfoldMultProj, and hUnfoldMultEventProj. Compare to external expectations or systematic variations, not to hTruthData.
Example	`matrixH("AO2D.root", true, "AO2Ddata.root", "AO2Dtruth.root", 1, 0, true)`

Closure / Data Modes

4. MC1 Response + MC2 Reco + MC2 Truth

Switches	useAO2DTraining = true, useAO2Data = 3
Meaning	trainAO2D = MC1 response/training. dataAO2D = MC2 reco-data-like input. truthAO2D = MC2 truth. centMin/centMax select the reco/data-like collisions only.
Compare	hUnfold to hTruthData. Projections: hUnfoldPtProj to hTruthDataPtProj, hUnfoldMultProj to hTruthDataMultProj, hUnfoldMultEventProj to hTruthDataMultEventProj. Ratios: hRatioPt, hRatioMult, hRatioMultEvent.
Caveat	With a non-inclusive centrality range, hTruthData remains inclusive unless the truth table also contains centrality. Use this mode for centrality closure only after adding a centrality-selected truth target.
Example	`matrixH("AO2D_MC1.root", true, "AO2D_MC2_data.root", "AO2D_MC2_truth.root", 3, 3, true)`

Closure / Data Modes

5. MC1 Response + MC2 NPMC Reco + MC2 NPMC Truth

Switches	useAO2DTraining = true, useAO2Data = 4
Meaning	trainAO2D = MC1 response/training. dataAO2D = MC2 O2npmcchargedtabl; reco pseudo-data comes from fPtRec/fMultNTracksNP and truth target from fPtGen/fMultGen. truthAO2D is not used.
Compare	hUnfold to hTruthData. Projections: hUnfoldPtProj to hTruthDataPtProj, hUnfoldMultProj to hTruthDataMultProj, hUnfoldMultEventProj to hTruthDataMultEventProj. Ratios: hRatioPt, hRatioMult, hRatioMultEvent.
Caveat	This mode does not fill true collision-level hMultRecoData, because O2npmcchargedtabl is row-based rather than a collision table.
Example	`matrixH("AO2D_MC1.root", true, "AO2D_MC2.root", "AO2Dtruth.root", 4, 4, true)`